Session 20

Implementing Enhancements in Existing Modules

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Objectives

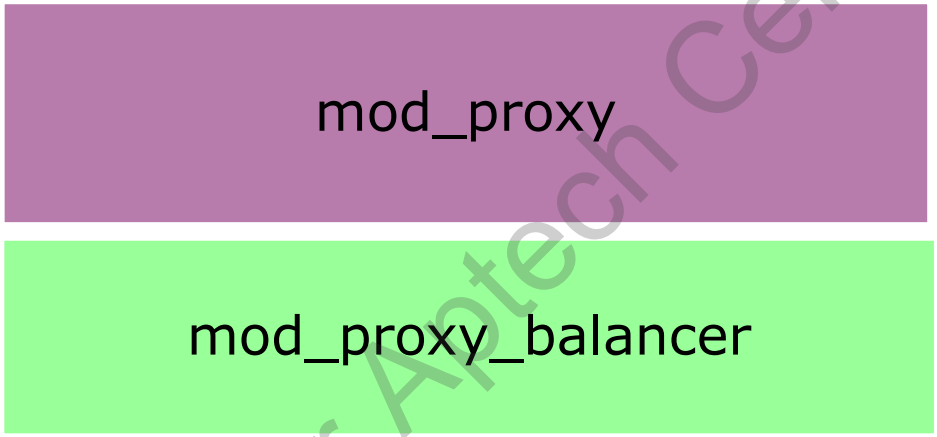
- Explain enhancements of the proxy modules
- Describe the enhancements in the Cache modules
- Outline the enhancement in the Cross-Site-Scripting modules
- Explain enhancements in the authentication and authorization modules

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Using the Proxy Modules

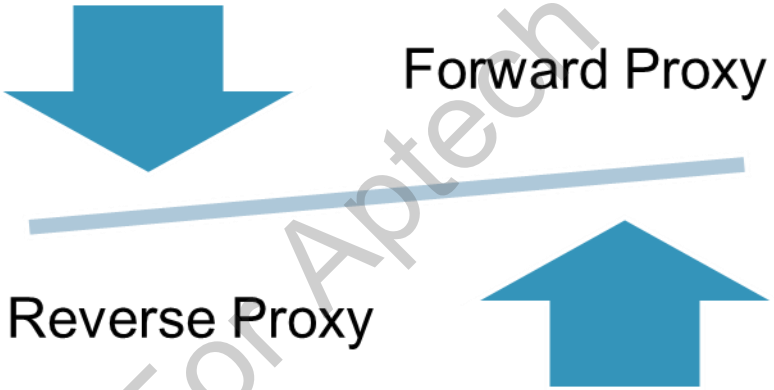
- Incorporates the improvement of the proxy server by providing an optimal performance to connect to a computer.
- Additional functionalities of the following modules are incorporated in Apache Web Server 2.4:



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Building a Proxy Server – mod_proxy

- Builds a proxy server for Apache Web Server.
- Works with the support of numerous load balancing algorithms and several other modules, such as `mod_proxy_fcgi`, `mod_proxy_scgi`, `mod_proxy_wstunnel`, `mod_proxy_http`, and so on.
- Modes of configuring Apache Web Server:



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Building a Proxy Server – Forward Proxy 1-2

- A central server is placed between the client and the source server.
- To receive content from the source server, the client directs a Web request to the central server (forward proxy) to further send it to the destination, which is the source server.
- Receives the content from the source server and sends it to the client.
- Provides Internet usage to the clients that are limited by a firewall.



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Building a Proxy Server – Forward Proxy 2-2

- Uses the `mod_cache` module for the caching capability of saving the network consumption.
- Sets the value of the `ProxyRequests` directive to On for configuring forward proxy.
- Example of a forward proxy:

```
ProxyRequests On
  ProxyVia On
  <Proxy *>
   Require host localhost:800
   </Proxy>
```

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Building a Proxy Server – Reverse Proxy 1-2

- No extra configuration is required in reverse proxy and is also known as a gateway.
- Acts like a normal Web server for a client and the client sends a Web request to the server assigned as a reverse proxy.
- Provides Internet usage to the clients that are behind the firewall.
- Provides caching to a slower database server so that load balancing can happen among a number of database servers.

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Building a Proxy Server – Reverse Proxy 2-2

- Uses the `ProxyPass` directive or the `RewriteRule` directive to configure the reverse proxy.
- Configures the source address of the proxy requests and also provides a support for UNIX.
- Example of a reverse proxy:

```
ProxyPass          "/xoo"
"http://localhost:800/car"
   ProxyPassReverse "/xoo" "http://
localhost:800/car"
```

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Performing Load Balancing - mod_proxy_balancer 1-2

Supports various protocols, such as File Transfer Protocol (FTP), HyperText Transmission Protocol (HTTP), WebSocket, and Apache JServ Protocol (AJP13) to perform load balancing.

Requires the load balancing scheduler algorithm to perform load balancing, but the `mod_proxy_balancer` module does not support this algorithm.

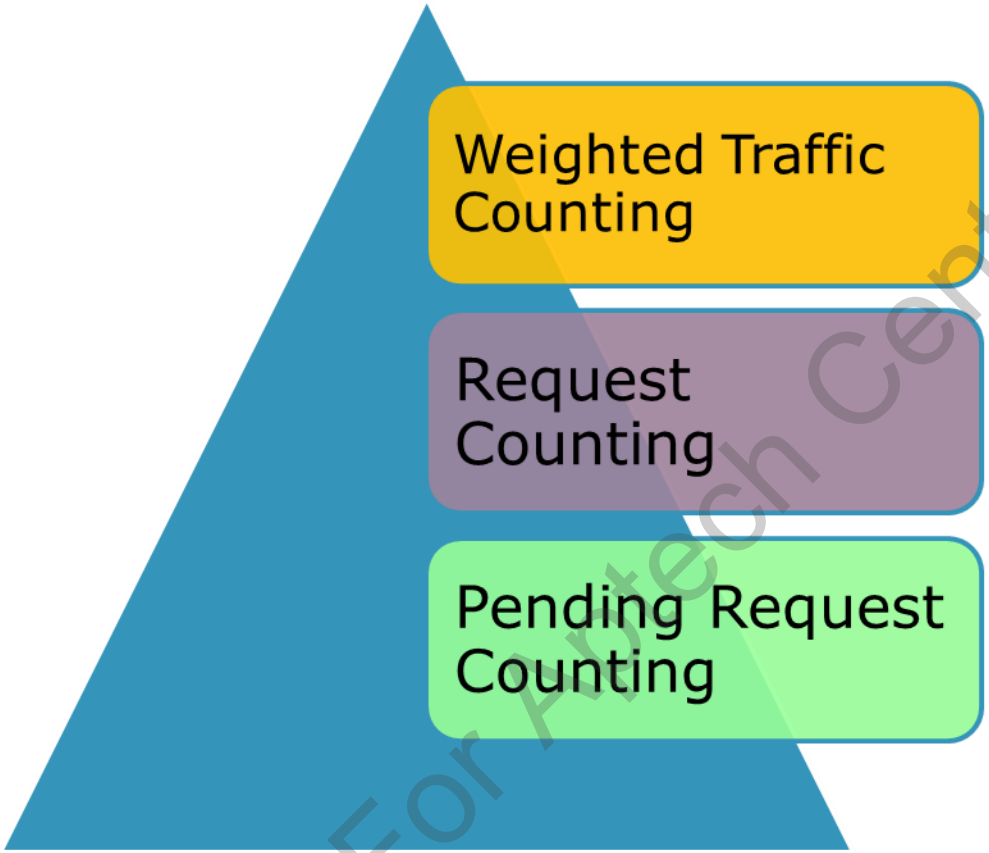
Performs load balancing using any one of the modules with the mod_proxy and mod_proxy_balancer modules:

- `mod_lbmethod_bytraffic`
- `mod_lbmethod_heartbeat`
- `mod_lbmethod_byrequests`
- `mod_lbmethod_bybusyness`

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- Load balancer scheduler methods:

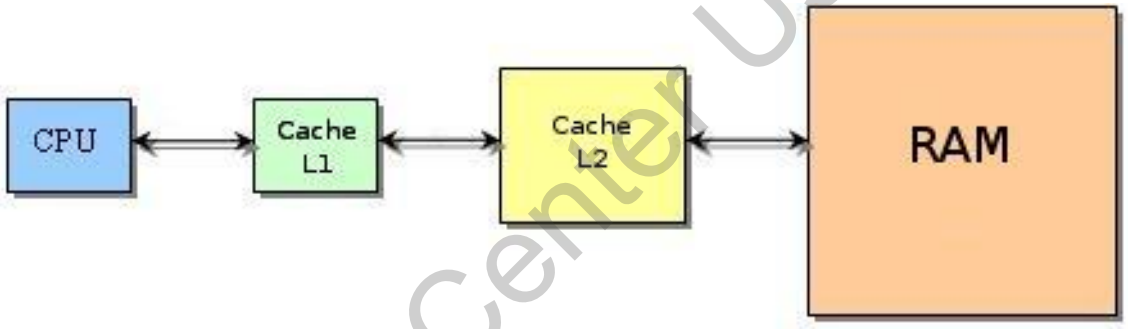




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Using the Cache Module

- Incorporates the improvement of caching functions in storing the frequently used data at a place where it can be retrieved easily.



- Additional functionalities of these modules are incorporated in Apache Web Server 2.4:

mod_cache

mod_info

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Caching Web Content – mod_cache 1-2

Using the Vary header defined in the Request for Comment (RFC) 2616.

Acts as a filter with content generated from any of the headers, including output of a CGI script, content proxied from other servers, flat flies, or dynamic content generator.

Keeps the caching filter at the top of the filter stack and avoids the processing of each request while retrieving content from the client using the quick handler.

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Caching Web Content – mod_cache 2-2

mod_cache_disk

- Works as a storage manager where content is stored on a disk, under a directory that is derived from the md5 hash of the cached URL.

mod_cache_socache

- Works as a storage manager where content is stored as a shared object cache.

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Example: Configure Cache in the httpd.conf File

- Configure cache in the httpd.conf file:

```
 LoadModule cache_module modules/mod_cache.so
   <IfModule mod_cache.c>
        LoadModule cache_disk_module
modules/mod_cache_disk.so
      <IfModule mod_cache_disk.c>
         CacheRoot c:/cacheapptech
         CacheEnable disk/
         CacheDirLevels 6
         CacheDirLength 4
     </IfModule>
   </IfModule>
```

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Enhancements in mod_cache

- Caches HEAD requests and is used in the filter chain to support proper control of caching.
- Configures the directives of `mod_cache` on a per directory basis instead of the per server basis.
- Customizes the cached URLs to provide the same endpoint URL prefix to a cluster of caches.
- Contains HIT/MISS/REVALIDATE in the X-Cache header that performs caching of the expired data if a server is unavailable.

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Providing Details of the Server Configuration – mod_info

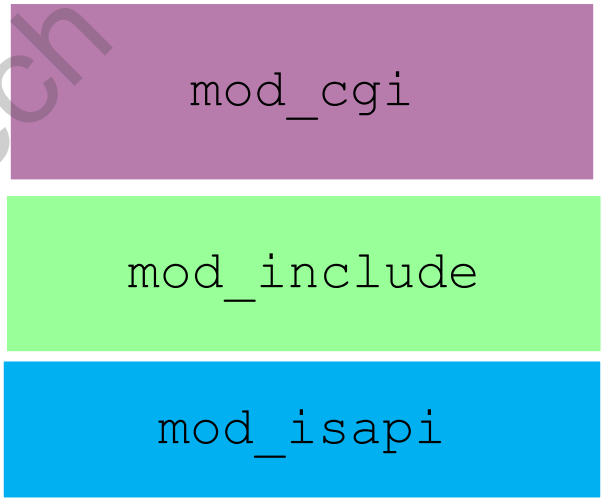
- Displays a detailed configuration of the server.
- Dumps the pre-parsed configuration to standard output while starting the server.
- Derives the pre-parsed configuration of server when the result of the `<IfModule>` and `<IfDefine>` directives are assessed based on the conditions and then the environment variables are changed with actual values.
- Example, to configure a detailed configuration of the server, update httpd.conf file with following code:

```
<Location "/server-info">
          SetHandler server-info
</Location>
```

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Using the Cross-Site-Scripting Attacks via Header Injection Modules

- Incorporates improvements to overcome the cross-site-scripting attacks via header injection.
- Resolves this problem by strictly following the rule of translating header to environment variables.
- Additional functionalities of these modules are incorporated in Apache Web Server 2.4.



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Executing CGI Scripts – mod_cgi

- Runs CGI script on Apache Web Server 2.4.
- Methods of specifying CGI script files are:
  - Using the `AddHandler` directive in the httpd.conf file.

```
AddHandler cgi-script .cgi
```

  - Including the path of the CGI script file in the `ScriptAlias` directive.

```
ScriptAlias "/cgi-bin/" "/web/cgi-bin/"
```

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Implementing Server-side Includes – mod_include

- Implements server-side includes filters.
- Uses Standard Generalized Markup Language (SGML) elements to parse HTML documents before sending to the clients.
- Uses the `INCLUDES` filter to implement server-side includes in httpd.conf file:

```
AddType text/html .shtml
AddOutputFilter INCLUDES .shtml
```

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Implementing Internet Server API Extensions – mod_isapi

- Uses the `mod_isapi` module to implement Internet Server Application Programming Interface (API) extensions.
- Runs the Internet Server extensions, such as .dll files for the Windows operating system only.
- For example, to implement this feature, you need to update the httpd.conf file and relate the ISAPI:

```
AddHandler isapi-handler .dll
```

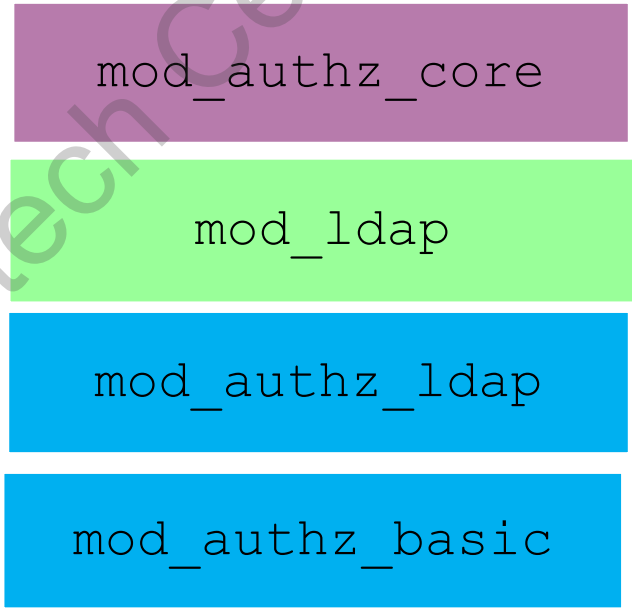
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Using the Authorization and Authentication Modules

- Incorporates improvement of authentication and authorization modules by using Lightweight Directory Access Protocol (LDAP) for controlling timeouts and debugging information.

- Additional functionalities of these modules are incorporated in Apache Web Server 2.4.



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Controlling Access – mod_authz_core

- Restricts or allows access to a Web site using the authorization logic.
- Helps to register several authorization providers using the `mod_authn_file` and `mod_authz_user` modules.
- Uses the `Require` directive with various authorization container directives, such as `<RequireNone>`, `<RequireAny>`, and `<RequireAll>` to specify the authorization logic.





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Example: An Authorization Logic 1-2

```
<Directory "/www/aptech">
    <RequireAll>
        <RequireAny>
            Require user superadmin
            <RequireAll>
                Require group admins
                Require ldap-group
"cn=Administrators,o=Airius"
                <RequireAny>
                    Require group HR
                    Require ldap-attribute dept="HR"
                </RequireAny>
            </RequireAll>
        </RequireAny>
```

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Example: An Authorization Logic 2-2

```
<RequireNone>
            Require group temps
            Require ldap-group "cn=Adhoc
Employees,o=Airius"
        </RequireNone>
    </RequireAll>
</Directory>
```

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Enhancing Performance – mod_ldap

- Enhances the performance of Web sites that are dependent on LDAP server for the backend connectivity.
- Provides an additional functionality of the shared memory cache and connection pool of LDAP server with the basic features of standard LDAP libraries.
- To implement these features, the `mod_authnz_ldap` module is used with its conjunction for enhancing the performance of HTTP Basic authentication.

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Example - To Enable the Connection Pool and Shared Memory Cache

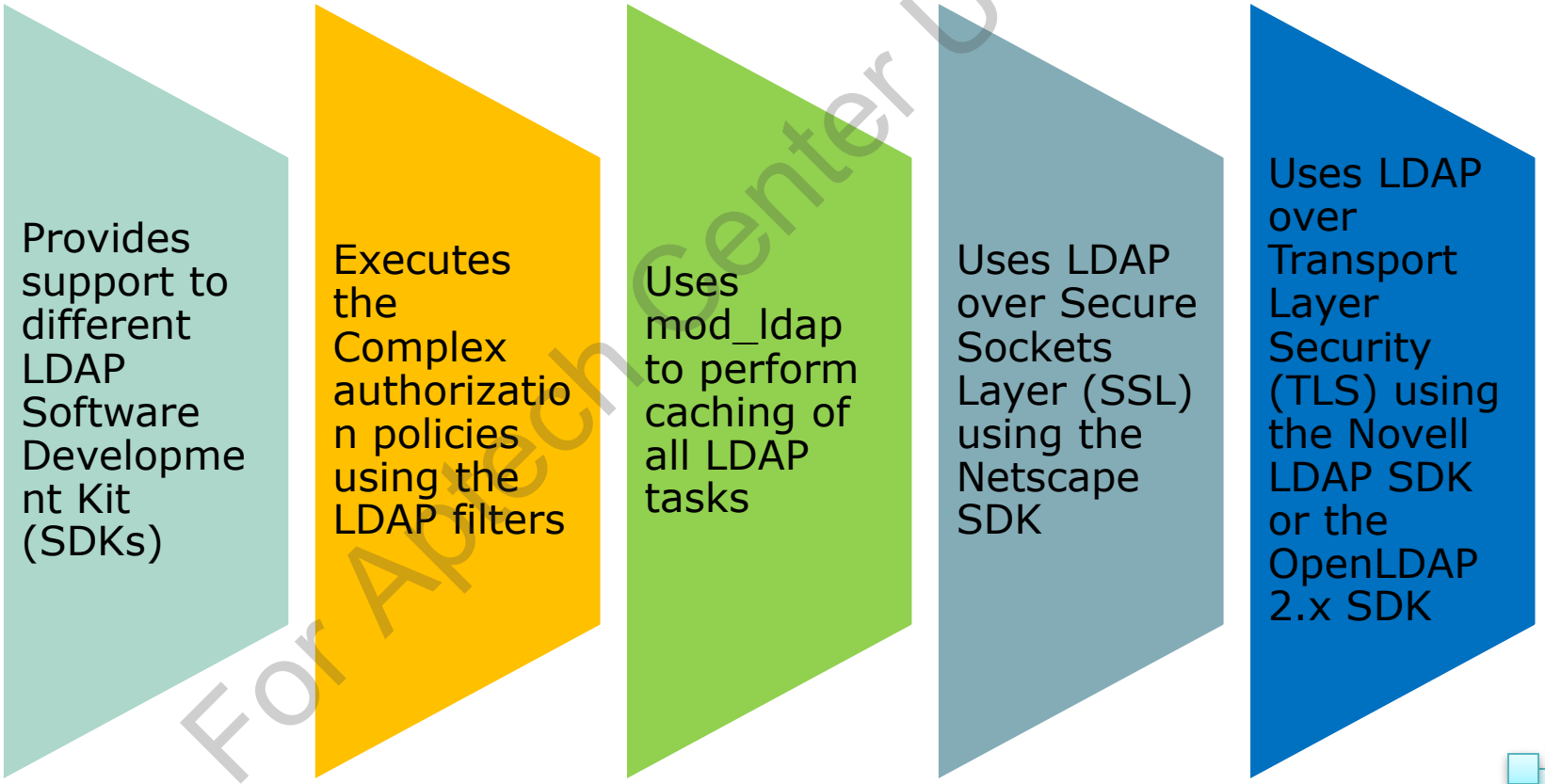
```
LDAPSharedCacheSize 600000
LDAPCacheEntries 1024
LDAPCacheTTL 700
LDAPOpCacheEntries 1024
LDAPOpCacheTTL 700
<Location "/ldap-status">
  SetHandler ldap-status
  Require host apptech.com
  Satisfy any
  AuthType Basic
  AuthName "LDAP Protected"
  AuthBasicProvider ldap
  AuthLDAPURL
ldap://localhost:800/dc=apptech,dc=com?uid?one
  Require valid-user
</Location>
```

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Validating Users - mod_authnz_ldap

- Validates users of `mod_auth_basic` using the LDAP directory.
- Supported features:



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Supporting Authentication Methods - mod_auth_basic 1-2

- Supports the HTTP Basic authentication method for restricting users to access Web resources after verifying their details in the authentication provider.
- Works with the combination of one of the modules of authentication modules, such as `mod_authn_file` and one of the authorization modules, such as `mod_authz_user`.
- Uses the `AuthBasicFake` directive to disable fake basic authentication specified in an Authorization header.

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Supporting Authentication Methods - mod_auth_basic 2-2

- The authorization header consists of two fields: `username` and `password` that are set based on the request parameters.
- To deactivate the false basic authentication, value of `AuthBasicFake` is set to Off.
- Syntax to use `AuthBasicFake` directive is:

```
AuthBasicFake off|username [password]
```

- For example, to specify a fixed `username` and `password` to an `AuthBasicFake` directive:

```
<Location "/aptech">
        AuthBasicFake dummy dummytest
</Location>
```

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Summary

- The mod_proxy module supports the ProxyPass directive that helps to simply to configure the server inside a LocationMatch or a Location directory for the optimal performance.
- The mod_proxy_balancer module supports various protocols, such as File Transfer Protocol (FTP), HyperText Transmission Protocol (HTTP), WebSocket, and Apache JServ Protocol (AJP13) to perform load balancing using the services of the mod_proxy module.
- You can use the mod_cache module as a filter with content generated from any of the headers, including output of a CGI script, and so on.
- The mod_info module is used to show a detailed configuration of the server.

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Summary

- The mod_isapi module is used to implement Internet Server Application Programming Interface (API) extensions.
- You can use the mod_authz_core module to restrict or allow access to a Web site using the authorization logic.
- You can use the mod_ldap module to enhance the performance of Web sites that are dependent on LDAP server for the backend connectivity.
- You can use the mod_authnz_ldap module to validate users of mod_auth_basic using the LDAP directory.
- You can use the mod_auth_basic module to support the HTTP Basic authentication method for restricting users to access the Web resources after verifying their details in the authentication provider.

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